Project Report

1. Student Details

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2. Project Details

- **Project Title**: Influencer Engagement and Sponsorship Coordination Platform (IESCP)
- **Problem Statement**: Develop a web-based application to facilitate collaboration between sponsors and influencers. The platform allows sponsors to create campaigns and manage ad requests while enabling influencers to view and respond to these requests.
- **Approach**:
    1. Designed and created database with segregation of users by role and separate tables for resources (campaigns and ad requests).
    2. Implemented role-based access control for users.
    3. Implemented key features such as campaign management, ad requests, user approval, flagging functionality, and registration.
    4. Used Celery for task scheduling (e.g., reminders) and asynchronous tasks, and SMTPLib for emailing.

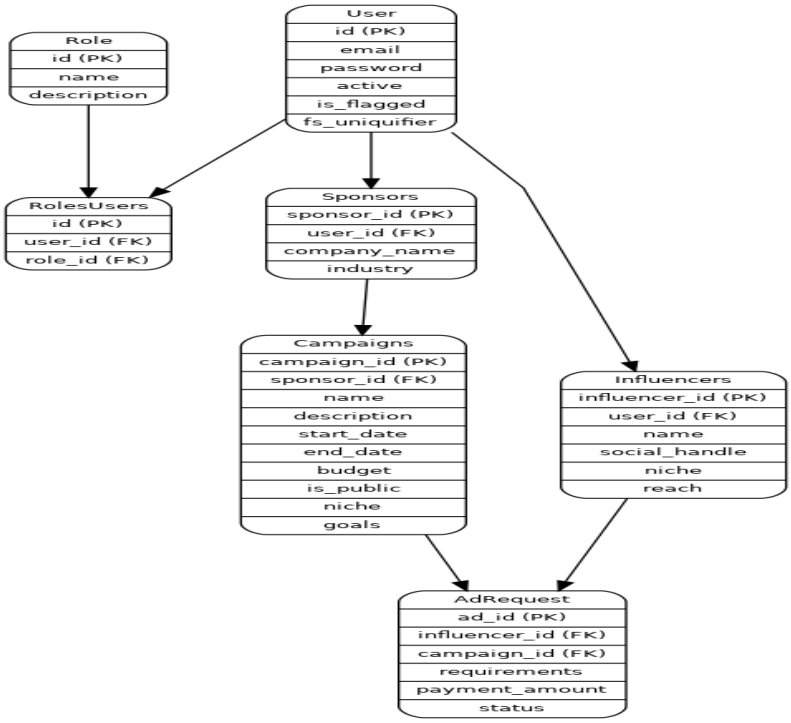
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3. Frameworks and Libraries Used

- **Frontend**:
    - Vue.js: For reactive UI components.
    - Bootstrap: For styling and responsiveness.
- **Backend**:
    - Flask: For building the application server.
    - Flask-SQLAlchemy: For ORM and database management.
    - Flask-Security: For authentication and user management.
    - SMTPLib: For email notifications.
    - Celery: For asynchronous tasks and scheduled jobs.
- **Database**:
    - SQLite.
- **Other Tools**:
    - Redis: As a message broker for Celery.
    - MailHog: For email testing during development.

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4. ER Diagram



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5. API Resource Endpoints

Endpoint	Method	Description
/campaign/<int:campaign_id>	GET	Retrieves details of a specific campaign.
/campaign/<int:campaign_id>	PUT	Updates a specific campaign (accessible by sponsors).
/campaign/<int:campaign_id>	DELETE	Deletes a specific campaign (accessible by sponsors).
/campaign	GET	Retrieves all campaigns.
/campaign	POST	Creates a new campaign (accessible by sponsors).
/sponsor-campaigns/<int:sponsor_id>	GET	Retrieves all campaigns created by a specific sponsor.
/campaign/<int:campaign_id>/toggle-visibility	PUT	Toggles the visibility of a campaign to public (accessible by sponsors).
/campaign-search	GET	Searches campaigns based on name or niche (only for public campaigns).
/ad	GET	Retrieves all ad requests.
/ad	POST	Creates a new ad request (accessible by sponsors).
/ad/<int:ad_id>	GET	Retrieves details of a specific ad request.

Endpoint	Method	Description
/ad/<int:ad_id>	PUT	Updates a specific ad request (accessible by sponsors).
/ad/<int:ad_id>	DELETE	Deletes a specific ad request (accessible by sponsors).
/requests/<int:influencer_id>	GET	Retrieves all ad requests assigned to a specific influencer.
/sponsor-ads	GET	Retrieves all ad requests created by a sponsor.
/admin-stats	GET	Provides admin statistics for active users, campaigns, and ad requests.
/ad-request/<int:ad_id>/status	PUT	Updates the status of an ad request to "Accepted" or "Rejected" (influencers).
/ad-request/<int:ad_id>/negotiate	PUT	Updates the payment amount of an ad request and sets status to "Pending Negotiation" (influencers).

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6. Presentation Video

- **Drive Link**: https://drive.google.com/drive/folders/1gURBNxr9_9xYZVIkX4SbjtBJTc4OJdiE?usp=sharing